Siddhant Surajlal Thalal

Research Assistant
I3D lab - IISc Bangalore
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Education

Indian Institute of Technology Jodhpur (IIT Jodhpur) 2021 – 2025
B.Tech in Mechanical Engineering | Minor in Robotics
CGPA: 8.01/10
Relevant Coursework: Introduction to Robotics, Control of Mechanical Systems, Cyber-Physical Systems, Mobile Robotics, Experimental Robotics, Autonomous Systems, Mechatronics

Research Experience

•**Research Assistant - I3D Lab, Indian Institute of Science (IISc)** *Nov 2025 – Present*
Supervisor: Prof. Pradipta Biswas

- Designed a tendon-driven, variable-length 13-DOF hyper-redundant aerial manipulator to expand UAV workspace flexibility.
- Developed a tendon-space kinematic formulation capturing coupled prismatic extension and multi-axis joint deflection constraints.
- Built an XR-based digital twin teleoperation framework integrating Unity, OptiTrack, and WebSockets for human-in-the-loop control.
- Led experimental validation on gravity-suspended prototype.
- Co-First author on IROS submission.

•**Research Intern, Indian Institute of Science (IISc)** *Aug 2025 – Nov 2025*
Supervisor: Prof. Pradipta Biswas

- Investigated hybrid wheeled–legged rover dynamics during stair descent.
- Reduced peak rear-wheel impact forces from 935 N to 615 N through fifth-leg load redistribution.
- Validated stable body articulation and expanded field of view via IMU-based experiments.

•**Camel-Inspired Quadruped Foot for Granular Locomotion** *Jan 2025 – May 2025*
Supervisor: Dr. Riby Abraham Boby - IIT Jodhpur

- Developed a camel-inspired biomimetic quadruped foot for sandy terrain locomotion using terramechanics principles.
- Designed an expanding paw mechanism to increase ground contact during stance and reduce sinkage (5mm) in granular media.
- Conducted sandbox experiments with ArUco tracking and ground-reaction-force sensing, achieving vertical body lift of 28.8mm vs. 8.1mm for a conventional rigid foot.
- Patent filed; IROS 2026 manuscript submitted as first author.

Selected Technical Projects

•**Underwater Robot with Buoyancy-Based Depth Regulation**
Undergraduate Thesis - Supervisor: Dr. Jayant Kumar Mohanta - IIT Jodhpur Aug 2025 – Nov 2025

- Designed a syringe-actuated buoyancy system with Arduino and pressure sensors for precise underwater depth control.
- Developed a PID feedback loop for precise depth control and integrated a Bluetooth app for real-time monitoring and data collection
- Validated stable depth modulation through simulations and controlled environment testing, achieving reliable underwater performance.

•**ROBOGUIDE – Autonomous Campus Navigation Robot**
(Coursework Project - Introduction to Robotics) Jan 2025 – April 2025

- Designed and built an autonomous robotic car integrating GPS, GSM, and ultrasonic sensors for real-time path planning and dynamic obstacle avoidance.
- Developed embedded software to process GPS data, execute motor control, and respond to remote user commands via GSM, enabling seamless end-to-end autonomous navigation.
- Achieved successful demonstrations of remote navigation through mapped environments with robust hardware–software integration and mobile app control.

•**LAND ROVER, JAGUAR - ROBOTIC CHARGING CHALLENGE**
(INTER IIT- TECHMEET 11.0) Jan 2023 – Feb 2023

- Developed ArUco-based vision system to localize charging port and autonomously align the connector.
- Implemented coordinate estimation and trajectory generation via inverse kinematics

Patents & Publications

•**Hyper-Redundant Aerial Manipulator: Tendon-Space Modeling and Experimental Validation**
Indian Institute of Science (IISc), Bangalore
Co-First Author; IROS 2026

•**Camel-Inspired Quadruped Foot Design for Granular Locomotion**
Indian Institute of Technology Jodhpur
First Author; Patent Filed: Camel-Inspired Biomimetic Foot for Granular Terrain Locomotion; IROS 2026

•**Design and Development of a Pitch-Enabled Leg–Wheel Mobile Robot**
Indian Institute of Science (IISc), Bangalore
Co-Author; IROS 2026

Conferences

AIR 2025 – 7th International Conference on Advances in Robotics *By the Robotics Society of India*
2–5 July 2025 IIT Jodhpur Conference-Website

- Developed and deployed an archival web platform on the IIT Jodhpur network for live demonstration at AIR 2025, later showcased by the supervising professor to the Robotics Society (TRS) of India as its official archive website.
- Engaged in advanced technical sessions, keynote addresses, and industry/startup showcases, gaining insights into cutting-edge robotics and AI research.
- Networked with leading researchers and industry professionals, strengthening collaborations and understanding of emerging trends in AI, automation, and embedded systems.

Technical Skills

- **Robotics Algorithms & Perception**: ROS 2, SLAM (2D), Path Planning, Motion Planning, Control Algorithms, Sensor Fusion
- **Programming**: Python, C++, MATLAB, Java, JavaScript, C
- **Data & Analytics**: Pandas, NumPy, Basic SQL, Power BI, SciPy, Matplotlib, Scikit-learn
- **Software & Simulation**: Unity, Matlab, Simulink, CAD (SolidWorks, AutoCAD, Onshape)
- **Tools & Frameworks**: Git, GitHub

Positions of Responsibility

•**Assistant Head**, INAE Youth Conclave, IIT Jodhpur *Sept 2022*